

Nathan Jacobson

COMPUTER SCIENCE STUDENT · DEVELOPER

✉ njacobson604@gmail.com 🏠 <https://nathanjacobson.ca> 🌐 grqphical 🌐 Nathan Jacobson

Experience

Tae Kwon Do Instructor

Port Moody, BC

TRANSCENDING MARTIAL ARTS

September 2025 - Present

- Instructed children aged 4-14 martial arts techniques such as strikes, blocks, patterns, and weapons training
- Lead a group of assistant instructors to aid with instruction and managing the class

Information Officer

Port Moody, BC

ELECTIONS CANADA

April 2025

- Confirmed voter identification
- Directed voter traffic and assisted voters with physical disabilities
- Recorded vote counts after polls had closed

Projects

intellidoc

🌐 grqphical/intellidoc

PYTHON, FASTAPI, SQLITE, LANCEDB

November 2025 - Present

- Currently developing a semantic search web application using Google's EmbeddingGemma model with both vector and keyword retrieval strategies, employing recursive chunking and metadata filtering to maximize accuracy.

interchange

🌐 grqphical/interchange

Go

August 2025-Present

- Created a reverse proxy in Golang that supports proxying, static file hosting, HTTPS, and IP blacklisting/whitelisting
- Currently working on adding rate limiting and load balancing

Remote Control and Autonomous Robotics System

🌐 grqphical/VRC-Code-2023

C++

September 2022 - February 2023

- Programmed both the autonomous and remote control software for my VEX VRC robotics team
- Robot was programmed to push discs into goal zones around the arena as well as roll a wall mounted roller to our team's colour
- Team placed 42nd in the province of British Columbia in autonomous skills competitions

Skills

Programming Languages

C, Python, Go, HTML, CSS, JS, SQL, Lua

Other Skills

Vim, Jupyter Notebooks, MS Office, Linux/WSL, Bash

Education

Simon Fraser University

Burnaby, BC

BSc IN COMPUTING SCIENCE

September 2025 - Present

Heritage Woods Secondary

Port Moody, BC

BC DOGWOOD DIPLOMA (HIGH SCHOOL DIPLOMA)

September 2021 - June 2025

- Awarded Top Producing Grade 12 Student
- Graduated with Honour Roll with Distinction

Extra-Curriculars

Tae Kwon Do

SPORT

September 2017 - Present

- Achieved a first degree black belt

Model United Nations

CLUB

September 2023 - June 2025

- Participated in multiple Model UN Conferences (CAHSMUN 2024, VMUN 2025, CAHSMUN 2025)
- Became co-president of my school's Model UN club

Certifications

BC FoodSafe Level One

March 2024